

## **SIMATS ENGINEERING**

### **ASSESSMENT TOOL 1**

**COURSE NAME: Database Management Systems**

**COURSE CODE: CSA0509**

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#### **COURSE OUTCOME COVERED**

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**CO1:** Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool: Short Answer Text (High)

Assessment Tool Weightage: 40%

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#### **Code of Conduct:**

I, Judy Allen J (192520055) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

#### **Signature of the student:**

Judy Allen J

**Q1. Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.**

A schema is the overall structure or blueprint of a database. It defines how data is organized, including tables, attributes, relationships, constraints, and data types. A schema is designed once and changes very rarely.

**Example:**

Student Table

Student\_ID Name Department

- The table structure (Student\_ID, Name, Department) is the schema.

**Instance**

An instance is the actual data stored in the database at a particular moment. Unlike the schema, an instance changes whenever records are inserted, updated, or deleted.

**Example:**

| Student_ID | Name  | Department |
|------------|-------|------------|
| 101        | Ravi  | CSE        |
| 102        | Priya | ECE        |

These records represent one instance of the Student table.

**Difference between Schema and Instance**

| Schema                         | Instance                           |
|--------------------------------|------------------------------------|
| Blueprint of the database      | Actual data stored in the database |
| Changes rarely                 | Changes frequently                 |
| Defines structure              | Contains records                   |
| Created during database design | Updated during database operations |

## **Types of Schema**

### **1. Logical Schema**

The logical schema describes the logical structure of the database. It defines tables, attributes, primary keys, foreign keys, relationships, and constraints without considering how data is physically stored.

#### **Example:**

Student(Student\_ID, Name, Department)

### **2. Physical Schema**

The physical schema describes how the data is actually stored on storage devices. It includes file organization, indexing methods, storage allocation, and access paths.

#### **Example:**

- B+ Tree Index on Student\_ID
- Data stored in disk blocks

### **3. View Schema**

A view schema provides customized views of the database for different users. It displays only the required data while hiding unnecessary or sensitive information.

#### **Example:**

A faculty member can view:

| Student\_ID | Name | Marks |

but cannot access student passwords or personal details.

## Comparison of Schemas

| Logical Schema               | Physical Schema                 | View Schema                           |
|------------------------------|---------------------------------|---------------------------------------|
| Describes database structure | Describes storage details       | Shows user-specific data              |
| Used by database designers   | Used by database administrators | Used by end users                     |
| Independent of storage       | Dependent on storage            | Independent of storage implementation |

## Conclusion

The schema defines the database structure, whereas the instance represents the actual data. The logical schema organizes data logically, the physical schema determines how data is stored, and the view schema provides customized access to users while maintaining security.

**Q2. Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.**

### **Entities and Attributes**

#### **1. Customer**

- Customer\_ID (Primary Key)
- Name
- Address
- Phone\_Number
- Email
- Date\_of\_Birth

#### **2. Account**

- Account\_Number (Primary Key)
- Account\_Type
- Balance
- Opening\_Date

#### **3. Branch**

- Branch\_ID (Primary Key)
- Branch\_Name
- Location
- IFSC\_Code

#### **4. Loan**

- Loan\_ID (Primary Key)
- Loan\_Type
- Loan\_Amount
- Interest\_Rate

## 5. Employee

- Employee\_ID (Primary Key)
- Name
- Designation
- Salary

## Relationships

### Customer → Owns → Account

- One customer can own many accounts.
- One account belongs to one customer.
- Relationship: One-to-Many (1:M)

### Branch → Maintains → Account

- One branch maintains many accounts.
- Relationship: One-to-Many (1:M)

### Branch → Employs → Employee

- One branch employs many employees.
- Relationship: One-to-Many (1:M)

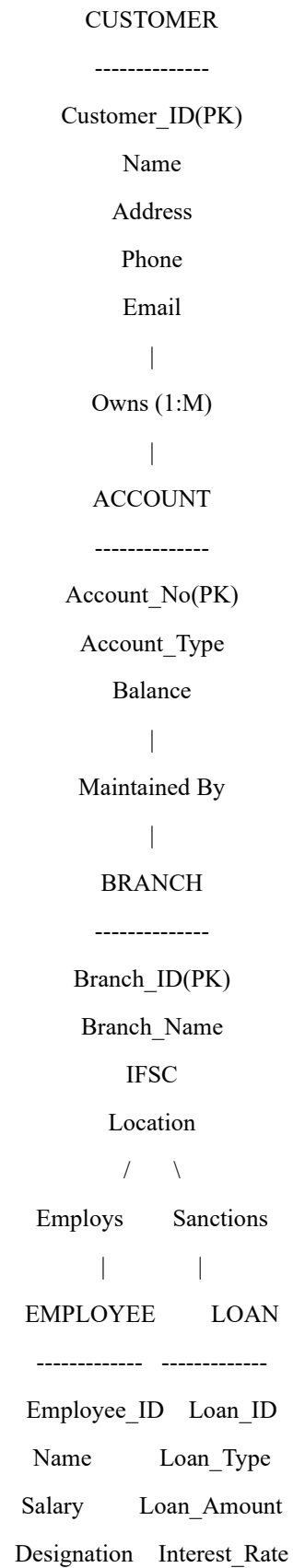
### Customer → Applies For → Loan

- One customer can apply for multiple loans.
- Relationship: One-to-Many (1:M)

### Branch → Sanctions → Loan

- One branch sanctions many loans.
- Relationship: One-to-Many (1:M)

## Textual ER Diagram



## **Conclusion**

The ER model represents the major entities of the Bank Management System along with their attributes and relationships. It helps in designing an efficient and normalized database.

**Q3. Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.**

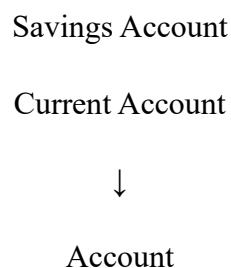
### **Extended ER (EER) Model**

The Extended Entity Relationship (EER) model is an advanced version of the ER model. It extends the traditional ER model by introducing inheritance, superclass-subclass relationships, specialization, generalization, and aggregation. These concepts help represent complex real-world applications more effectively.

#### **1. Generalization**

Generalization is the process of combining multiple lower-level entities with common characteristics into a single higher-level entity called a superclass.

#### **Example**

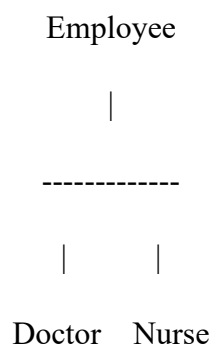


Both Savings Account and Current Account share common attributes such as Account Number and Balance, so they are generalized into the Account entity.

#### **2. Specialization**

Specialization is the opposite of generalization. It divides a higher-level entity into multiple lower-level entities based on specific characteristics.

#### **Example**

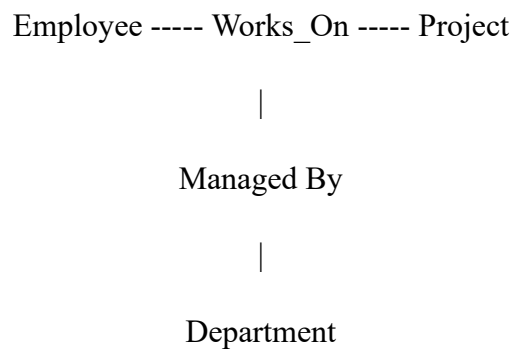


Doctor and Nurse inherit all common Employee attributes while having their own specialized attributes.

### 3. Aggregation

Aggregation is used when a relationship itself needs to participate in another relationship. It treats a relationship as a higher-level entity.

#### Example



Here, the relationship **Works\_On** is aggregated and linked to Department.

#### Advantages of EER Model

- Represents complex real-world scenarios.
- Supports inheritance and reuse of attributes.
- Reduces redundancy.
- Improves database organization.
- Makes database design more flexible and scalable.

#### Conclusion

The EER model enhances the ER model by introducing generalization, specialization, and aggregation. These concepts improve database modeling, reduce duplication, and accurately represent complex relationships.

**Q4. Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.**

### **Entities**

#### **Patient**

- Patient\_ID (PK)
- Name
- Age
- Gender
- Address

#### **Employee (Superclass)**

- Employee\_ID (PK)
- Name
- Salary
- Phone

#### **Doctor (Subclass)**

- Specialization
- Qualification

#### **Nurse (Subclass)**

- Shift
- Experience

#### **Department**

- Department\_ID (PK)
- Department\_Name

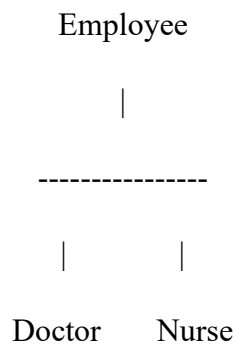
#### **Appointment**

- Appointment\_ID (PK)
- Date
- Time

#### **Medical\_Record (Weak Entity)**

- Record\_No (Partial Key)
- Diagnosis
- Prescription

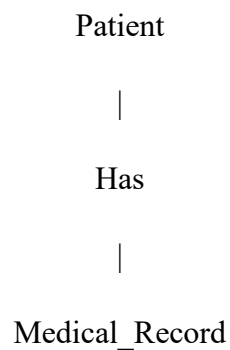
## Inheritance (Specialization)



Doctor and Nurse inherit Employee attributes while having their own specialized attributes.

## Weak Entity

Medical\_Record is a weak entity because it cannot exist without a Patient.



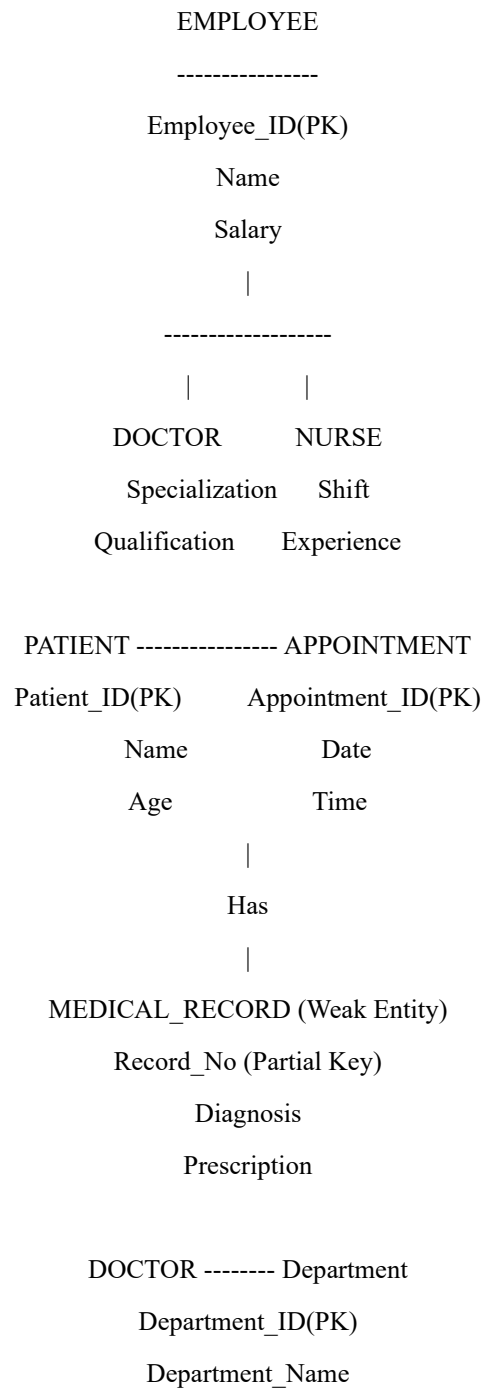
The primary key of Medical\_Record is formed using:

Patient\_ID + Record\_No

## Relationships

- Patient books Appointment (1:M)
- Doctor attends Appointment (1:M)
- Doctor belongs to Department (M:1)
- Nurse works in Department (M:1)
- Patient owns Medical\_Record (1:M)

## Textual EER Diagram



## Conclusion

The Hospital Management System uses specialization to model Doctor and Nurse as subclasses of Employee and uses Medical\_Record as a weak entity dependent on Patient. This design improves database organization and reduces redundancy.

**Q5. Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.**

A Database Administrator (DBA) is a professional responsible for managing, maintaining, securing, and optimizing a database system. The DBA ensures that the database is available, reliable, efficient, and protected from unauthorized access.

**Responsibilities of a DBA**

**1. Database Installation and Configuration**

- Install DBMS software.
- Configure database settings.
- Create and manage databases.

**2. Database Design**

- Define schemas, tables, indexes, and relationships.
- Ensure proper normalization and efficient database structure.

**3. Security Management**

- Create user accounts.
- Assign roles and privileges.
- Protect data from unauthorized access.

**4. Backup and Recovery**

- Perform regular database backups.
- Restore data in case of system failure.
- Implement disaster recovery plans.

**5. Performance Tuning**

- Optimize SQL queries.
- Create indexes.
- Monitor database performance and improve response time.

## **6. Data Integrity**

- Enforce constraints.
- Maintain consistency and accuracy of stored data.

## **7. Concurrency Control**

- Manage simultaneous access by multiple users.
- Prevent conflicts using transactions and locking mechanisms.

## **8. Database Monitoring**

- Monitor storage usage, memory, CPU utilization, and logs.
- Detect and resolve issues before they affect users.

## **9. Software Maintenance**

- Apply patches and software updates.
- Upgrade DBMS versions while ensuring compatibility.

## **10. Documentation**

- Maintain documentation for database design, security policies, backup procedures, and recovery plans.

## **Conclusion**

The DBA plays a crucial role in ensuring the database remains secure, reliable, efficient, and available. Their responsibilities include installation, design, security, backup, recovery, performance tuning, monitoring, and maintenance, making them essential for the successful operation of any database system.